1. Create a directory named lab6 and a text file named permissions.txt inside it.

```
juhee@kali[~]  $ mkdir lab6
juhee@kali[~]  $ cd lab6
juhee@kali[~/lab6]  $ touch permissions.txt
juhee@kali[~/lab6]  $ ls -l

total 0
-rw-rw-r-- 1 juhee juhee 0 Nov  2 14:38 permissions.txt
```

2. Display the long listing of permissions.txt and identify the owner, group and current rwx permissions.

```
juhee@kali[~/lab6]  $ ls -l permissions.txt

-rw-rw-r-- 1 juhee juhee 0 Nov  2 14:38 permissions.txt
```

3. Change permissions using symbolic notation so owner has read, write and execute; group has read and write; others have no permissions.

```
juhee@kali[~/lab6]  $ chmod u=rwx,g=rw,o= permissions.txt
juhee@kali[~/lab6]  $ ls -l permissions.txt

-rwxrw---- 1 juhee juhee 0 Nov  2 14:38 permissions.txt
```

4. Change permissions using numeric mode 640 and verify the result.

```
juhee@kali[~/lab6]  $ chmod 640 permissions.txt
juhee@kali[~/lab6]  $ ls -l permissions.txt

-rw-r----- 1 juhee juhee 0 Nov  2 14:38 permissions.txt
```

5. Change permissions using numeric mode 755 and verify the result.

```
juhee@kali[~/lab6]  $ chmod 755 permissions.txt
juhee@kali[~/lab6]  $ ls -l permissions.txt

-rwxr-xr-x 1 juhee juhee 0 Nov  2 14:38 permissions.txt
```

6. Remove execute permission from the owner using symbolic chmod notation.

```
juhee@kali[~/lab6]  $ chmod u-x permissions.txt
juhee@kali[~/lab6]  $ ls -l permissions.txt

-rw-r-xr-x 1 juhee juhee 0 Nov  2 14:38 permissions.txt
```

7. Add read and execute permissions for the group without changing the owner's permissions.

```
juhee@kali[~/lab6]  $ chmod g+rx permissions.txt
juhee@kali[~/lab6]  $ ls -l permissions.txt

-rw-r-xr-x 1 juhee juhee 0 Nov  2 14:38 permissions.txt
```

8. Remove all permissions from others and verify the result.

```
juhee@kali[~/lab6]  $ chmod o= permissions.txt
juhee@kali[~/lab6]  $ ls -l permissions.txt

-rw-r-x--- 1 juhee juhee 0 Nov  2 14:38 permissions.txt
```

9. Create a directory named secureddata and set its permissions to 750.

```
juhee@kali[~]  $ mkdir secureddata
juhee@kali[~]  $ chmod 750 secureddata
juhee@kali[~]  $ ls -ld secureddata

drwxr-x--- 2 juhee juhee 4096 Nov  2 14:38 secureddata
```

10. Demonstrate the difference between file and directory read, write and execute permissions by testing access to a practice directory.

```
juhee@kali[~]  $ mkdir practice
juhee@kali[~]  $ touch practice/demo.txt
juhee@kali[~]  $ chmod 640 practice/demo.txt
juhee@kali[~]  $ chmod 750 practice
juhee@kali[~]  $ ls -ld practice
juhee@kali[~]  $ ls -l practice

drwxr-x--- 2 juhee juhee 4096 Nov  2 14:38 practice
-rw-r----- 1 juhee juhee 0 Nov  2 14:38 demo.txt

Directory: r = list names, w = create/delete entries, x = enter/search.
File: r = read contents, w = modify contents, x = execute.
```

11. Display the current umask value and record the result.

```
juhee@kali[~]  $ umask

0022
```

12. Temporarily change umask to 027, create a new file and directory, and verify their permissions.

```
juhee@kali[~]  $ umask 027
juhee@kali[~]  $ touch umask_file.txt
juhee@kali[~]  $ mkdir umask_dir
juhee@kali[~]  $ ls -l umask_file.txt
juhee@kali[~]  $ ls -ld umask_dir
```

```
-rw-r----- 1 juhee juhee 0 Nov  2 14:38 umask_file.txt
drwxr-x--- 2 juhee juhee 4096 Nov  2 14:38 umask_dir
```

13. Restore umask to its previous value and verify the current umask.

```
juhee@kali[~]  $ umask 022
juhee@kali[~]  $ umask
```

```
0022
```

14. Create a new user and group for ACL practice, then create aclfile.txt for testing.

```
juhee@kali[~]  $ sudo groupadd aclgroup
juhee@kali[~]  $ sudo useradd -m acluser
juhee@kali[~]  $ touch aclfile.txt
juhee@kali[~]  $ ls -l aclfile.txt
```

```
-rw-rw-r-- 1 juhee juhee 0 Nov  2 14:38 aclfile.txt
```

15. Display the current ACL of aclfile.txt using getfacl and identify the owner, group, user, group and mask entries.

```
juhee@kali[~]  $ getfacl aclfile.txt
```

```
# file: aclfile.txt
# owner: juhee
# group: juhee
user::rw-
group::rw-
other::r--
```

16. Give a specified user read and write permission on aclfile.txt without changing normal owner/group permission bits. Verify using getfacl.

```
juhee@kali[~]  $ sudo setfacl -m u:acluser:rw aclfile.txt
juhee@kali[~]  $ getfacl aclfile.txt
```

```
# file: aclfile.txt
# owner: juhee
# group: juhee
user::rw-
user:acluser:rw-
group::rw-
mask::rw-
other::r--
```

17. Give a specified group read permission on aclfile.txt and verify the ACL.

```
juhee@kali[~]  $ sudo setfacl -m g:aclgroup:r aclfile.txt
juhee@kali[~]  $ getfacl aclfile.txt
```

```
# file: aclfile.txt
# owner: juhee
# group: juhee
user::rw-
user:acluser:rw-
group::rw-
group:aclgroup:r--
mask::rw-
other::r--
```

18. Modify and remove a specific ACL entry from aclfile.txt. Verify using getfacl.

```
juhee@kali[~]  $ sudo setfacl -m u:acluser:r-- aclfile.txt
juhee@kali[~]  $ sudo setfacl -x g:aclgroup aclfile.txt
juhee@kali[~]  $ getfacl aclfile.txt
```

```
# file: aclfile.txt
# owner: juhee
# group: juhee
user::rw-
user:acluser:r--
group::rw-
mask::rw-
other::r--
```

19. Set a default ACL on a directory so new files inherit the required permissions. Create a test file and verify the inherited ACL.

```
juhee@kali[~]  $ mkdir acldir
juhee@kali[~]  $ sudo setfacl -m d:g:aclgroup:rx acldir
juhee@kali[~]  $ touch acldir/test.txt
juhee@kali[~]  $ getfacl acldir/test.txt
```

```
# file: acldir/test.txt
# owner: juhee
# group: juhee
user::rw-
group::r--
group:aclgroup:r--
mask::r--
other::r--
```

20. Create, modify, verify and clean up file permissions and ACLs using chmod, umask, setfacl and getfacl.

```
juhee@kali[~]  $ touch workflow.txt
juhee@kali[~]  $ chmod 640 workflow.txt
juhee@kali[~]  $ umask 022
juhee@kali[~]  $ sudo setfacl -m u:acluser:rw workflow.txt
juhee@kali[~]  $ getfacl workflow.txt
juhee@kali[~]  $ sudo setfacl -b workflow.txt
juhee@kali[~]  $ rm workflow.txt
juhee@kali[~]  $ sudo userdel -r acluser
juhee@kali[~]  $ sudo groupdel aclgroup
```

```
# file: workflow.txt
# owner: juhee
# group: juhee
user::rw-
user:acluser:rw-
group::r--
mask::rw-
other::---

ACL entries cleared; workflow file, test user and test group removed.
```